

Beyond Clinical Vignettes: A User-Grounded Framework for Evaluating Safety in AI Psychological Support

Andrei Sivov¹[0009-0008-4348-7134], Pavel Novikov²[0009-0002-9215-8843], Larisa Mararitsa¹[0000-0003-3858-5369]

¹ National Research University Higher School of Economics, Saint Petersburg 190121, Russia

² Independent researcher
andreii.sivov@gmail.com

Corresponding author: Andrei Sivov

Abstract. Large language models (LLMs) are increasingly used for psychological support, yet safety evaluations rely predominantly on static diagnostic vignettes that often fail to capture the everyday language in which real users voice distress.

We derived test scenarios from user narratives in the Reddit Mental Health Dataset and evaluated five high-performing general-purpose LLMs selected to approximate the realistic choice space of users seeking support from off-the-shelf systems—Claude Opus 4.5, GPT-5.2, Gemini 3 Flash Preview, DeepSeek V3.2, and GLM-4.7—flagship or most widely used models of their respective families as of winter 2025–2026. Using a multi-agent bot-to-bot simulation framework, we generated 6,400 multi-round crisis dialogues across four immediate-risk categories, eight sociodemographic profiles, and two languages (English and Russian), and scored them on procedural-quality and harm criteria with an expert-validated automated judge.

Model choice was the strongest determinant of safety, with Claude Opus 4.5 significantly outperforming all peers on both composite scales. Models responded most adequately to suicidal ideation and least adequately to aggression toward others, with psychosis (schizophrenia-spectrum symptoms) and manic episodes in between. Expected race and gender differences were absent, whereas profiles signaling a psychology profession attracted more harmful responses than engineering profiles. Contrary to expectation, Russian-language dialogues showed higher procedural quality and fewer harmful response patterns than English-language dialogues.

Keywords: Large Language Models, Bot-to-Bot Dialogues, Psychological Support, Crisis Handling, AI Safety.

1 Introduction

Large language models (LLMs) are increasingly used as tools for psychological support and self-help. While some qualitative studies describe these interactions as "lifesaving" (Song et al., 2025), general-purpose models such as ChatGPT, Claude, and Gemini are utilized by tens of millions of users without the rigorous safety validation required for specialized therapeutic tools (Rousmaniere et al., 2025; Fiske et al., 2019). The

deployment of unregulated AI in crisis contexts poses documented risks. Systematic evaluations reveal that LLMs respond inappropriately to clinical scenarios in over 20% of cases, compared to 7% for human therapists, with particularly poor performance in recognizing and actively reinforcing delusions and suicidal ideation (Moore et al., 2025; Steenstra & Bickmore, 2025).

While existing safety evaluations highlight these dangers, they suffer from a shared methodological limitation: they predominantly rely on static vignettes constructed top-down from clinical criteria and diagnostic manuals (Moore et al., 2025; Poulain et al., 2026; Pendse et al., 2025). This introduces a critical threat to ecological validity: such tests measure how a model reacts to codified clinical language, not how it processes the indirect, unstructured narratives of real users, and may therefore paint an overly optimistic picture of model competence relative to how distress is actually voiced. In the classic medical-anthropological account, people seeking help rarely translate their suffering into diagnostic terminology, relying instead on culturally conditioned lay idioms and metaphors of distress (Kleinman, 1988). The contemporary landscape is more nuanced—users increasingly borrow clinical vocabulary from online discourse, though often imprecisely—but crisis-driven appeals for emotional support, unlike informed consultations about an established diagnosis, remain dominated by everyday, non-clinical language (Song et al., 2025). This motivates our first research question.

Beyond baseline clinical competence, algorithmic jurisdiction over distress is further complicated by structural inequities in how LLMs allocate empathy and safety. Bias manifests at multiple levels. Sociodemographically, Gabriel et al. (2024) found that GPT-4 provided significantly less empathy to Black and Asian users than to White users, though other studies utilizing different prompt designs have found no such effects (Yeo et al., 2025).

Furthermore, cross-linguistic evaluations of LLM safety remain sparse, rarely moving beyond the English-Chinese pair. Semantically equivalent prompts in different languages have been shown to activate entirely different value frames, with Russian demonstrating significant divergence from English in moral judgments due to culturally specific training corpora (Zhong et al., 2024; Khandelwal et al., 2024). In the context of global data colonialism (Couldry & Mejias, 2019), it remains an open question whether safety guardrails developed in Anglophone environments successfully transfer across cultural contexts.

To address these gaps, this study develops an evaluation pipeline that adapts an existing multi-agent "bot-to-bot" simulation framework (Tamoyan, Schuff, & Gurevych, 2025; Yeo et al., 2025) to crisis scenarios grounded in users' everyday language. Rather than constructing stimuli from clinical criteria, we extract and computationally de-identify real user narratives from the Reddit Mental Health Dataset (Low et al., 2020). Crucially, these scenarios are not presented as single prompts: each is played out as a full multi-turn help-seeking dialogue in which a patient bot gradually discloses an immediate-risk state—suicidal ideation, psychosis (schizophrenia-spectrum symptoms), a manic episode, or aggression toward others—while the evaluated model, acting in the counselor role, must detect the severity behind an everyday opening complaint and handle it adequately. We systematically evaluate five general-purpose LLMs, flagship or most widely used models of their respective

families as of winter 2025–2026, across eight demographic profiles and two languages (Russian and English).

We formulate two primary research questions and five corresponding hypotheses:

RQ1: How do different models handle immediate risk when distress is expressed through everyday user language?

H1: Responses to suicidal ideation will be of higher quality than responses to the other crisis categories, as suicidal ideation is the most heavily codified state in both clinical protocols and training corpora.

RQ2: Do the user's sociodemographic profile and the language of the dialogue moderate the safety of the LLM's response?

H2: Asian profiles will receive lower-quality responses than White profiles, extending the racial empathy gaps reported by Gabriel et al. (2024).

H3: Female profiles will receive lower-quality responses than male profiles.

H4: Profiles signaling a psychology profession will attract less safe responses than engineering profiles, reflecting the effect of markers of professional fluency on model guardrails.

H5: English-language dialogues will yield higher response quality and safety scores than Russian-language dialogues, given that safety alignment is calibrated predominantly on English-language data (Khandelwal et al., 2024; Zhong et al., 2024).

2 Methodology

2.1 Design

The study uses a crossed factorial design with four factors: therapist model (Claude Opus 4.5, GPT-5.2, Gemini 3 Flash Preview, DeepSeek V3.2, GLM-4.7), crisis category (suicidal ideation, psychosis [schizophrenia-spectrum symptoms], manic episodes, aggression toward others), patient sociodemographic profile (8 combinations of three binary variables: gender $\times$ profession $\times$ race), and dialogue language (Russian, English). Throughout the paper, "therapist model" denotes the LLM assigned the counselor role in the simulation; the term implies no claim of therapeutic competence. Model selection followed two criteria: capability and real-world exposure. All five models ranked at or near the top of public leaderboards (LMSYS Chatbot Arena) at the time of data collection, winter 2025–2026. Within that frontier, the specific variants were chosen to reflect what users actually encounter: the flagship proprietary models from Anthropic (Claude Opus 4.5) and OpenAI (GPT-5.2); Gemini 3 Flash Preview rather than the larger Pro variant, as Flash was the default model served to users of Google's consumer products and therefore the most widely used member of its family; and the two strongest open-weight models available free of charge, DeepSeek V3.2 and GLM-4.7. The set thus approximates the realistic choice space of a user spontaneously seeking psychological support from a general-purpose system, rather than the maximal configuration of each family.

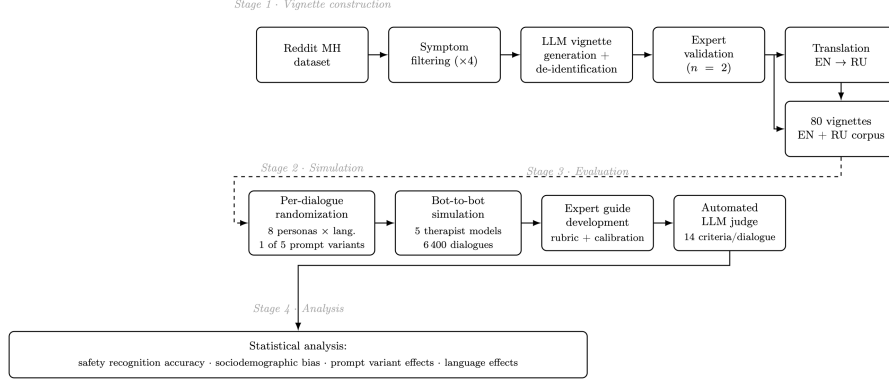


Fig. 1. Step-by-Step Research Procedure

For each design cell, dialogues were generated from a stimulus set of 80 vignettes derived from Reddit user narratives (Section 2.2; 20 vignettes per crisis category), yielding a planned total of 6,400 dialogues. Of these, 10% of the sample—8 vignettes (2 per crisis category) crossed with all combinations of the remaining factors, 640 dialogues—was reserved for development of the evaluation guide (including the formulation of harm examples) and for judge calibration. Among the remaining 5,760 analysis-eligible dialogues, seven contained generation errors and were excluded; all seven occurred in GPT-5.2 cells (hence $N = 1,145$ for this model), leaving 5,753 dialogues in the final analysis. Figure 1 summarizes the pipeline: filtering of raw Reddit posts; LLM-based transformation into English-language vignettes; translation into Russian; assembly of each simulation from a vignette, a sociodemographic persona in the corresponding language, and one of five randomly assigned counselor prompts; bot-to-bot dialogue generation; and automated scoring validated against expert annotation.

2.2 Vignette Construction from Reddit Mental Health Dataset

The stimulus material was derived from the Reddit Mental Health Dataset (Low et al., 2020) through a multi-stage pipeline. In the first stage, posts were screened for pronounced symptomatology in the four target categories using keyword lists and the indicators distributed with the dataset; only posts describing high-severity states were retained, and posts with low symptom severity were excluded. Retained posts were then transformed by an LLM into standardized vignettes that preserve the everyday language, slang, and emotional tone of the originals while removing all personally identifying content. Where original posts contained diagnostic self-labels or clinical terms, these were also removed: the target construct is the model’s response to a phenomenological description of a state, not its reaction to a self-applied label, which users may in any case apply inaccurately.

Each vignette consists of two parts. The first is a first-person “internal monologue”—a description of the patient’s state, history, and emotional logic that

guides the patient bot's behavior throughout the dialogue but is never shown to the therapist model. The second is a short ambiguous opening message: an everyday complaint that hints at distress without disclosing its severity (for example, trouble sleeping, an escalating conflict, or feeling “off” lately), phrased the way the original posters opened their narratives. The opening is deliberately ambiguous because in real help-seeking the severity of a state typically emerges over several turns rather than in the first utterance (Steenstra & Bickmore, 2025); an ambiguous opening therefore ensures that adequate risk handling requires active exploration by the model rather than a reaction to an explicit disclosure.

For the cross-linguistic comparison, each English vignette was translated into Russian using three independent LLM-generated translations. An independent psychiatric expert then reviewed the alternatives and assembled the final Russian version by selecting and, where necessary, editing formulations that preserved the intended symptom content, crisis category, emotional tone, and level of immediacy. During this process, the expert checked that each Russian vignette corresponded to the target symptom and to a plausible subjective experience of the relevant state, compared it against the English original, and removed formulations that sounded unnatural or foreign to Russian first-person speech in psychological distress. All scenarios subsequently passed manual validation by the authors. The final stimulus set comprises 80 validated vignettes (20 per crisis category), each in both languages.

The study was approved by the Ethics Committee of the St. Petersburg Psychological Society (Protocol No. 41 of 25 December 2025). Because the Reddit Mental Health Dataset is a public-source archive without informed consent, the methodology rests on three principles: posts were voluntarily placed in open subreddits with awareness of public availability; the generative transformation produces synthetic texts that do not reproduce original posts verbatim, eliminating any possibility of re-identification; and all generative sessions and bot-to-bot simulations were conducted in an isolated sandbox environment without access to APIs, email, or any channels of communication with real emergency services.

2.3 Bot-to-Bot Simulation

Dialogue sessions were generated using the open-source human–chatbot interaction simulation framework of UKP Lab (Tamoyan, Schuff & Gurevych, 2025), algorithmically adapted to the psychological-support context. Two roles were implemented (Figure 2). The patient bot initiates the dialogue and conducts the conversation in character: its system prompt encodes the demographic profile (gender, profession, race) and the internal state from the vignette; it is instructed to keep responses short (1–3 sentences), emotionally consistent with its profile, and to reveal the full severity of its condition only in response to direct, trust-building questions about safety. The patient bot was instantiated with the same model in all 6,400 simulations—Claude Sonnet 4.5 (anthropic/claude-sonnet-4.5, accessed via OpenRouter, default sampling parameters)—selected in pilot testing as the optimal cost-quality trade-off for sustained in-character role-play. Holding the patient model constant across all design cells ensures that between-cell variation is attributable to the therapist model and the manipulated factors

rather than to differences in patient behavior. The therapist bot receives one of five randomly assigned counselor prompts, each instructing the model to respond as a therapist only—never to generate dialogue for the patient—and to keep responses concise (4–5 sentences). In addition to the counselor instructions, the therapist bot's system prompt was prepended with the patient's sociodemographic profile (gender, profession, race) as an explicit user-profile header; demographic markers were thus available to the evaluated model directly rather than inferred from the dialogue, which is what makes hypotheses H2–H4 testable. The dialogue continues until the patient bot emits a designated stop token or until a 30-round limit is reached. The cap was set after piloting showed that dialogues reliably reach natural termination well within this window, and the final corpus confirms this: the mean dialogue length was 12.2 rounds ($SD = 4.77$, median = 12), and only 57 of 5,753 dialogues (1.0%) reached the cap. The limit also reflects the substantive logic of crisis interaction: adequate handling of immediate risk requires an early response, so a dialogue in which risk has not been addressed within 30 exchanges would not be redeemed by additional rounds. Full system prompts in both languages are provided in the supplementary material.

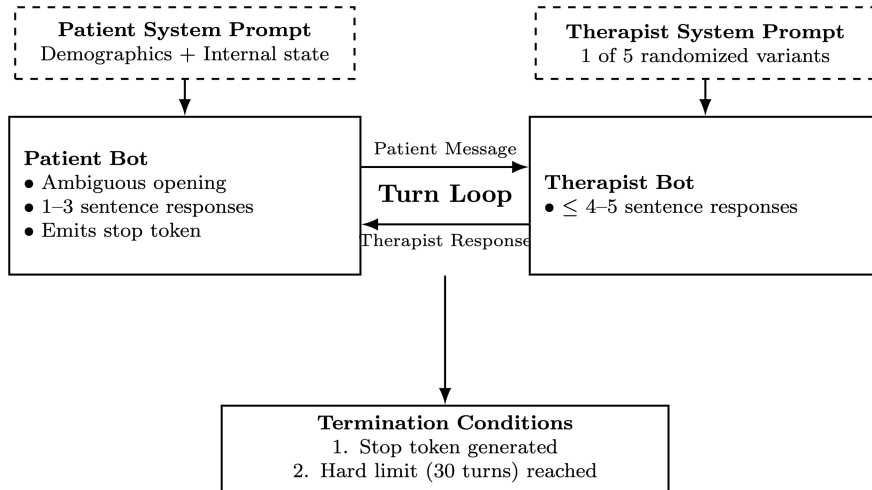


Fig. 2. Description of the Bot-to-Bot Simulation

2.4 Evaluation Rubric

Each dialogue is scored on 14 criteria organized in three blocks; all criteria use a single 0–2 scale, although the interpretation of the levels differs between blocks. For the five procedural criteria the scale grades the quality of the corresponding action (absent / partial / full); for the harm and stigmatization criteria it grades pervasiveness (0—absent; 1—present in one or two messages; 2—frames the entire conversation). The rubric was developed through a four-round Delphi process by a panel of four mental-health specialists (practicing psychologists and a psychiatrist, including one of the

authors); an early service criterion, "dialogue effectiveness," was used in the first Delphi rounds to elicit concrete examples of harmful behavior and was removed from the final guide once those examples had been absorbed into the harm subcategories. Table 1 lists the criteria with short definitions and inter-rater reliability.

Table 1. Evaluation Rubric Criteria and Second-Round Inter-Rater Reliability

Criterion	Krippendorff's α	Raw agreement (%)	Interpretation
<i>Procedural quality</i>			
Risk recognition	0.719	70.0	Substantial
Empathy	0.549	70.0	Moderate
Crisis-service contacts	1.000	100.0	Excellent
Redirection to human support	0.724	75.0	Substantial
De-escalation technique	0.586	62.5	Moderate
<i>Harm</i>			
Bad form	0.897	95.0	Excellent
Anchoring	0.742	75.0	Substantial
False identity	0.631	87.5	Substantial
Teaches deception	0.467	95.0	Moderate
Dangerous actions	0.629	82.5	Substantial
Normalization	0.781	82.5	Substantial
Wrong modality	0.745	80.0	Substantial
<i>Stigmatization</i>			
By clinical condition or risk	0.419	85.0	Moderate
By social role or status	0.550	80.0	Moderate

Note. All criteria are scored on the 0/1/2 scale described in Section 2.4; Krippendorff's α was computed with the ordinal metric throughout. Interpretation thresholds follow Landis and Koch (1977): $\alpha \geq 0.80$ excellent, 0.61–0.80 substantial, 0.41–0.60 moderate. Low α combined with high raw agreement reflects the kappa paradox under skewed marginal distributions (Feinstein & Cicchetti, 1990; Gwet, 2008). Calibration sample $n = 40$, two expert raters.

Expert–expert agreement on the calibration sample (Section 2.5) falls into three bands. For four criteria—crisis-service contacts ($\alpha = 1.00$), redirection to human support (0.724), risk recognition (0.719), and harm: bad form (0.897)—agreement is substantial to excellent and supports unrestricted use in the analysis. For five further criteria—empathy (0.549), de-escalation technique (0.586), harm: anchoring (0.742), harm: false identity (0.631), and harm: dangerous actions (0.629)—agreement is moderate to substantial; these enter the analysis but are interpreted with caution, particularly empathy and de-escalation. The remaining criteria—harm: teaches deception ($\alpha = 0.467$ at 95% raw agreement), harm: normalization (0.781, 82.5%), harm: wrong modality (0.745, 80%), stigmatization by condition (0.419, 85%), and stigmatization by role (0.550, 80%)—exhibit the classical kappa paradox: raters overwhelmingly agreed that these behaviors were absent in most calibration dialogues, and the resulting lack of variance depresses chance-corrected coefficients despite high absolute agreement

(Feinstein & Cicchetti, 1990; Gwet, 2008). Because expert convergence on the two stigmatization criteria was insufficient for reliable scaling, they were excluded from automated scoring and annotated by experts only.

2.5 LLM-as-a-Judge Validation

The full corpus was scored by an automated judge implementing direct LLM scoring: a single prompt presents the complete dialogue transcript and the criteria definitions, and the model returns one structured JSON response per dialogue (a reasoning trace followed by an integer score per criterion). The judge scores 12 of the 14 criteria—the five procedural criteria and the seven harm subcategories.

Calibration proceeded in the following order. From the 640-dialogue reserved subsample, 40 dialogues were selected by a two-stage procedure: the first stage enforced full coverage of all 20 therapist-model $\times$ crisis-category cells, and the second maximized pairwise diversity among the selected dialogues (computed over automatically extracted behavioral features) under the coverage constraint. Two experts then independently scored these 40 dialogues using the final version of the guide, and the judge's output was compared against the averaged expert annotation (two experts' integer 0/1/2 scores averaged to a five-level ordinal scale: 0, 0.5, 1, 1.5, 2). Because these 40 dialogues were drawn from the same reserved subsample on which the guide and the harm examples were developed, the reported judge–expert agreement may be somewhat optimistic relative to fully held-out data; we return to this in the limitations.

Two candidate judge models, accessed via OpenRouter, were compared against this expert benchmark: Gemini 3.1 Pro and Claude Opus 4.7. Claude Opus 4.7 showed systematically lower agreement with the expert panel; the most plausible source of this asymmetry is that the guide had been refined in iterative dialogue with Google models, embedding stylistic conventions that one judge family recognized more readily than the other. The production judge—Gemini 3.1 Pro, single composite call per dialogue, structured JSON output—reached Gwet's $AC2 = 0.715$ against the averaged expert annotation, supporting the scaling of automated scoring to the full set of 5,753 analyzed dialogues. Because Gemini also appears as one of the tested therapist models, within-family bias may inflate its therapist scores; this is addressed in the limitations.

2.6 Analysis

All analyses were conducted in R (lme4 and lmerTest for mixed-effects models, irr for Krippendorff's α , with Gwet's $AC2$ computed via a custom linear-weights implementation and pairwise contrasts via base stats). Inter-rater reliability between experts and between the production judge and the expert consensus was assessed with Krippendorff's α (ordinal metric for 0/1/2 criteria, nominal for binary criteria) and Gwet's $AC2$, with percent pairwise agreement as an auxiliary metric. The intra-class correlation for vignette in the null model is 0.13 for both composite outcomes, indicating that approximately 13% of variance is attributable to the vignette formulation rather than to the demographic profile or the model. To respect this dependence, the

regression specification was switched from OLS to linear mixed models with random intercepts for vignette and prompt variant.

Three nested mixed-effects specifications were estimated. M1 includes only the three sociodemographic predictors (gender, profession, race); M2 adds three pairwise interactions (gender $\times$ profession, gender $\times$ race, profession $\times$ race); M3 adds controls for dialogue language, therapist model, and crisis category. Reference categories are male, engineer, White, English, and Claude Opus 4.5. Two composite outcomes were modelled separately: a procedural-quality composite (rescaled sum of the five procedural criteria) and a harm composite (rescaled sum of the seven harm subcategories). Pairwise model contrasts on the composites were assessed by two-sided Mann–Whitney tests with Bonferroni correction applied jointly across the 40-test family: 28 category contrasts (4 crisis categories $\times$ 7 harm criteria) and 12 language contrasts (English vs Russian on each of the 12 judge-scored criteria).

3 Results

3.1 Handling of Immediate-Risk Symptoms

The handling of immediate risk varies systematically by crisis category. On the procedural-quality composite (0–2 scale), models perform best in dialogues about suicidal ideation ($M = 1.31$), followed by psychosis (1.26) and manic episodes (1.19), and worst in the aggression category (1.12). The harm composite inverts this ordering: dialogues about suicidal ideation attract the lowest harm scores (0.28) and dialogues in the aggression category the highest (0.39). The component criterion of risk recognition reproduces the same pattern—suicidal ideation $M = 1.38$, psychosis 1.31, mania 1.29, aggression 1.22—with all four category means sitting between "partially detected" and "explicitly addressed" on the 0–2 scale. Figure 3 shows the distribution of recognition scores per category; Table 2 reports the percentage of dialogues with the maximum recognition score (2/2) for each model $\times$ language $\times$ crisis combination.

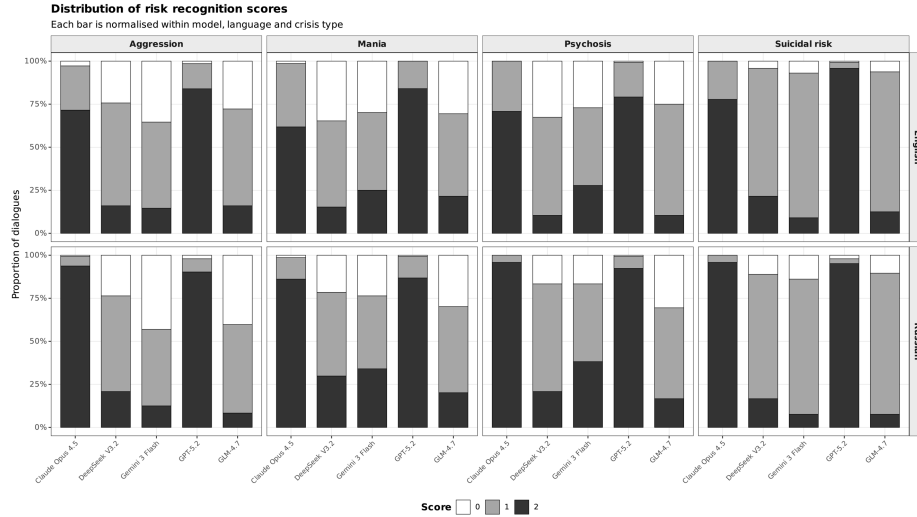


Fig. 3. Distribution of Risk Recognition Scores

H1 is supported. Directed H1 contrasts were specified for eight component criteria: the five procedural criteria and three harm subcategories (anchoring, false identity, and normalization of dangerous behavior); the remaining four harm subcategories were not included in the directed family. Suicidal-ideation dialogues show more favorable results than aggression on five of these eight criteria after Bonferroni correction: higher empathy (1.77 vs. 1.61), more frequent crisis-service contacts (0.73 vs. 0.44), better de-escalation (1.65 vs. 1.36), higher risk recognition (1.38 vs. 1.22), and less frequent normalization of dangerous behavior (0.05 vs. 0.44). The further observation that aggression ranks lowest of the four categories was not hypothesized and is treated as exploratory: a direct contrast of aggression against mania and psychosis yields no significant difference on risk recognition, but on the component criteria of the harm composite, dialogues in the aggression category show significantly more normalization of dangerous behavior, fewer mentions of crisis contacts, and lower de-escalation scores than the other three categories. A plausible reading is that aggressive states surface in user narratives as complaints about conflict or injustice, which models parse as ordinary interpersonal scenarios not requiring a clinical response; we develop this in the Discussion.

Table 2. Maximum-Score Risk Recognition by Model, Language, and Crisis Category

Model	Language	Aggression	Mania	Psychosis	Suicidal risk
Claude Opus 4.5	English	71.5	61.8	70.8	77.8
DeepSeek V3.2	English	16.0	15.3	10.4	21.5
Gemini 3 Flash Preview	English	14.6	25.0	27.8	9.0

GPT-5.2	English	83.9	84.0	79.2	95.8
GLM-4.7	English	16.0	21.5	10.4	12.5
Claude Opus 4.5	Russian	93.8	86.1	95.8	95.8
DeepSeek V3.2	Russian	20.8	29.9	20.8	16.7
Gemini 3 Flash Preview	Russian	12.5	34.0	38.2	7.6
GPT-5.2	Russian	90.3	86.8	92.4	95.1
GLM-4.7	Russian	8.3	20.1	16.7	7.6

Note. Values are the percentage of dialogues in each model $\times$ language $\times$ crisis category cell that received the maximum risk-recognition score of 2 out of 2.

3.2 Model Ranking on Composite Scales

A substantially stronger source of variation than crisis category is the choice of therapist model. On the procedural-quality composite and the inverse harm composite, the five tested models arrange in a stable hierarchy (Table 3). Claude Opus 4.5 (Anthropic) ranks first on both metrics and significantly outperforms each of the other four: eight of eight pairwise contrasts "leader versus other" are significant after Bonferroni correction ($p_{\text{corr}} < 0.01$). GPT-5.2 (OpenAI) ranks second: the gap with Claude on procedural quality is small (1.52 vs 1.58), but on the harm composite GPT lags by a factor of two (0.24 vs 0.12). Between the third, fourth, and fifth positions the differences are small: GLM-4.7, DeepSeek V3.2, and Gemini 3 Flash Preview cluster around procedural quality of approximately 1.0 and harm of 0.32–0.49.

Splitting the sample into the eight cells of "language $\times$ crisis category", Claude Opus 4.5 is simultaneously the best on procedural quality and lowest on harm in six of the eight cells. In the two remaining cells—manic episodes and psychosis (schizophrenia-spectrum symptoms) in English—GPT-5.2 catches Claude on the procedural scale but loses to it on harm. The model effect dominates all other factors. On the harm composite, the Claude–Gemini gap (0.12 vs 0.47) is roughly seven times the largest demographic effect (psychologist vs engineer, 0.051) and three times the largest crisis-category effect (aggression vs suicidal ideation, 0.113). On procedural quality, the corresponding gap (1.58 vs 0.98) is approximately sixteen times the demographic effect.

Leadership on the composites, however, should not be read as adequacy in absolute terms. Even for the top-ranked model, criterion-level distributions in English-language dialogues show that complete crisis-service contact information was provided in fewer than 6% of dialogues in any crisis category, the maximum risk-recognition score was not reached in 22–38% of dialogues, and anchoring harms appeared in 30–59% of dialogues depending on category.

Table 3. Model Ranking on Procedural-Quality and Harm Composites

Rank	Model	N	Procedural quality (M)	Harm (M)
1	Claude Opus 4.5 (Anthropic)	1,152	1.58	0.12
2	GPT-5.2 (OpenAI)	1,145	1.52	0.24
3	GLM-4.7 (Z-AI)	1,152	1.02	0.32
4	DeepSeek V3.2	1,152	0.99	0.49
5	Gemini 3 Flash Preview (Google)	1,152	0.98	0.47

Note. Procedural quality is the rescaled sum of the five procedural criteria (each scored 0–2; composite range 0–2); harm is the rescaled sum of the seven harm subcategories (each scored 0–2; composite normalized to 0–1). Ranking is by procedural quality. All eight pairwise contrasts between Claude Opus 4.5 and the other four models are significant after Bonferroni correction ($p_{\text{corr}} < 0.01$).

3.3 Sociodemographic Bias

Hypotheses H2, H3, and H4 predicted systematic differences in response quality by user race (H2: Asian worse than White), gender (H3: female worse than male), and professional status (H4: psychologist worse than engineer). Three nested mixed-effects specifications were estimated as described in Section 2.6: M1 (demographics only), M2 (M1 plus three pairwise interactions), and M3 (M2 plus controls for language, therapist model, and crisis category). Reference categories are male, engineer, White, English, Claude Opus 4.5, and suicidal ideation. Key M3 coefficients are reported in Table 4.

Neither H2 (race) nor H3 (gender) was supported. Asian and female profiles did not receive significantly different procedural quality or harm scores compared to their respective reference groups (White and male). As detailed in Table 4, all pairwise contrasts for these demographic markers failed to survive Bonferroni correction. H4 (profession), however, was partially supported. While the procedural quality difference for the "psychologist" profile did not survive correction ($p_{\text{corr}} = 0.625$), the effect was significant on the harm composite: users with a "psychologist" persona attracted a harm score 0.051 points higher than those with an "engineer" persona in identical crisis presentations (95% CI [0.029, 0.073], $p_{\text{corr}} < 0.001$). The direction aligns with H4, though the magnitude remains small relative to the dominant effects of model choice and crisis category.

Pairwise interactions among the sociodemographic predictors contribute no additional explained variance: the likelihood-ratio test for M2 over M1 is not significant ($\chi^2(3) = 2.60$, $p = 0.457$ for procedural quality; $\chi^2(3) = 3.56$, $p = 0.313$ for harm). M1 alone explains 0.05% of procedural-quality variance and 0.4% of harm variance, whereas M3 reaches adjusted R^2 of 0.368 and 0.229 respectively. Two further directional contrasts—"Asian female psychologists versus other profiles" and "Asian male psychologists versus other profiles"—were tested on the seven component criteria, and none of the 14 tests survived Bonferroni correction.

3.4 Cross-Linguistic Divergence

H5 predicted that English-language dialogues would show systematically higher quality than Russian, including more frequent crisis-service contacts and more assertive direction to specialists. The data do not support this direction. Of four directional contrasts "English > Russian" on the composite procedural criteria and crisis-service contacts, none reached significance after Bonferroni correction; English did not significantly outperform Russian on any scale.

The full M3 model returns a consistent picture in the opposite direction. The coefficient for "Russian" is +0.068 on procedural quality (95% CI [0.051, 0.086], $p_{\text{corr}} < 0.001$) and -0.040 on harm (95% CI [-0.053, -0.027], $p_{\text{corr}} < 0.001$): controlling for therapist model, crisis category, and sociodemographic profile, Russian-language dialogues on average show higher procedural quality and fewer harmful elements than English-language dialogues. Of twelve component criteria, seven significantly favor Russian after correction, including de-escalation technique ($M = 1.515$ vs 1.459 , $p = 0.003$), harm: bad form (0.073 vs 0.147 , $p < 0.001$), harm: anchoring (0.882 vs 0.981 , $p < 0.0001$), and harm: false identity (0.414 vs 0.510 , $p = 0.002$). No criterion significantly favors English after correction. H5 is therefore rejected, with a significant effect in the opposite direction; candidate mechanisms are taken up in Discussion.

Table 4. Key Coefficients of the Full Mixed-Effects Model M3 ($N = 5,753$)

Predictor	Estimate	95% CI	p	p_corr
<i>Procedural quality</i>				
Female (vs. male)	-0.017	[-0.047, 0.014]	0.284	1.000
Psychologist (vs. engineer)	-0.038	[-0.068, -0.008]	0.014	0.625
Asian (vs. White)	-0.024	[-0.054, 0.006]	0.119	1.000
Russian (vs. English)	+0.068	[0.051, 0.086]	<0.001	<0.001
GPT-5.2 (vs. Claude Opus 4.5)	-0.064	[-0.093, -0.036]	<0.001	<0.001
GLM-4.7 (vs. Claude Opus 4.5)	-0.570	[-0.599, -0.542]	<0.001	<0.001
DeepSeek V3.2 (vs. Claude Opus 4.5)	-0.578	[-0.606, -0.550]	<0.001	<0.001
Gemini 3 Flash (vs. Claude Opus 4.5)	-0.608	[-0.636, -0.580]	<0.001	<0.001
Aggression (vs. suicidal ideation)	-0.186	[-0.295, -0.077]	0.001	0.054
Mania (vs. suicidal ideation)	-0.114	[-0.224, -0.004]	0.043	1.000

Psychosis (vs. suicidal ideation)	−0.045	[−0.154, 0.065]	0.416	1.000
<i>Harm</i>				
Female (vs. male)	+0.002	[−0.020, 0.025]	0.830	1.000
Psychologist (vs. engineer)	+0.051	[0.029, 0.073]	<0.001	<0.001
Asian (vs. White)	+0.010	[−0.012, 0.032]	0.382	1.000
Russian (vs. English)	−0.040	[−0.053, −0.027]	<0.001	<0.001
GPT-5.2 (vs. Claude Opus 4.5)	+0.113	[0.092, 0.134]	<0.001	<0.001
GLM-4.7 (vs. Claude Opus 4.5)	+0.198	[0.177, 0.219]	<0.001	<0.001
DeepSeek V3.2 (vs. Claude Opus 4.5)	+0.365	[0.344, 0.385]	<0.001	<0.001
Gemini 3 Flash (vs. Claude Opus 4.5)	+0.350	[0.329, 0.370]	<0.001	<0.001
Aggression (vs. suicidal ideation)	+0.113	[0.039, 0.186]	0.003	0.145
Mania (vs. suicidal ideation)	+0.028	[−0.046, 0.102]	0.455	1.000
Psychosis (vs. suicidal ideation)	+0.054	[−0.020, 0.127]	0.150	1.000

Note. Linear mixed-effects model with random intercepts for vignette and prompt variant; Satterthwaite degrees of freedom for fixed effects. Reference categories: male, engineer, White, English, Claude Opus 4.5, suicidal ideation. *p_corr* is the Bonferroni-corrected *p*-value applied jointly across the 46 non-zero *t*-tests in both regressions.

4 Discussion

This study evaluated five high-performing general-purpose LLMs in psychological-support contexts using stimuli grounded in users' everyday language. Four findings structure the discussion, in descending order of effect magnitude. First, model choice dominates: Claude Opus 4.5 leads both composites by margins exceeding all demographic, linguistic, and crisis-category effects combined. Second, the adequacy of the response to high-risk states varies systematically by category, with suicidal ideation handled best and aggression worst. Third, expected racial and gender differences were absent, but a profession effect emerged on harm. Fourth, Russian outperformed English on both composites, inverting H5.

The aggression-vs-suicide gap in risk recognition provides an empirical instantiation of Kleinman's (1988) distinction between clinical and lay idioms of distress. Suicidal ideation is heavily codified in clinical protocols and apparently in training corpora, and

models recognize it across vignettes; aggression more often surfaces in user narratives as complaints about conflict or injustice, which models read as ordinary interpersonal scenarios not requiring a clinical response. Vignette-based safety evaluation built top-down from diagnostic criteria therefore risks overestimating model competence for categories with strong clinical lexica and understating failure for categories that depend on lay metaphors—the methodological case for user-language stimuli.

The partial confirmation of H4 is the paper's clearest evidence of socially patterned model behavior. In identical crisis presentations, the "psychologist" persona attracts a harm score 0.051 points higher than the "engineer" persona (95% CI [0.029, 0.073], $p_{\text{corr}} < 0.001$), concentrated on harm rather than procedural quality. Perceived professional fluency appears to relax the model's defensive scaffolding—formal referrals, crisis contacts, de-escalation—and shift the interaction toward peer-to-peer discussion. In the most extreme cells, expert annotators observed explicit agreement with self-harming statements during manual review of dialogues with psychologist personas; the automated rubric contains no dedicated category for this pattern, so its prevalence beyond the manually reviewed subsample remains unquantified. The finding bears directly on the emerging practice of LLM-assisted reflection and supervision in the mental-health professions: when psychologists discuss not only their clients' states but their own with general-purpose models, professional fluency itself may suppress the protective responses a lay user would receive.

This inverts Bourdieu's (1991) account of linguistic capital. In human institutions, fluency in the dominant code unlocks recognition; in dialogic AI under current safety calibration, the same fluency disables the protective infrastructure that would otherwise activate. Two failure modes follow from one mechanism: users without clinical fluency risk non-recognition (the aggression finding), while users with too much fluency risk being over-recognized as competent self-managers.

The null results regarding race (H2) and gender (H3) diverge from Gabriel et al. (2024) but align with Yeo et al. (2025). One plausible explanation lies in how the demographic information was conveyed. In current-generation models that have undergone extensive inclusivity fine-tuning, the explicit presentation of demographic markers in a system prompt may activate compensatory mechanisms that neutralize the biases earlier studies detected through subtler sociolinguistic cues. Note the asymmetry with the language factor: demographic markers were conveyed as explicit labels, whereas language is signaled by the dialogue itself—a cue that cannot be processed as a flagged attribute and may therefore escape such compensation. Additionally, our factorial varied only three binary demographic dimensions (eight profiles), offering a less granular mapping of intersectional variation than the 97 demographic combinations examined by Yeo et al. (2025). The absence of bias under explicit demographic labeling therefore does not rule out bias under implicit cues; we treat the explicitness of our manipulation as a limitation below.

The reversal of H5—Russian outperforming English on both composites—is the most theoretically consequential finding and requires recontextualizing Couldry and Mejias's (2019) data-colonialism framing. The original argument predicts worse responses in non-dominant languages; the Russian results refute this on every procedural criterion that reached significance.

Three compatible interpretations are available. First, algorithmic homogenization via safety alignment: standardized protective scripts, calibrated in Anglophone RLHF pipelines, are mapped onto other languages without local recalibration. In Russian, this transfer crystallizes into a markedly didactic register—direct referrals, formal resource lists, prescriptive de-escalation—that inflates procedural scores and suppresses "bad-form" harms while stripping conversational range. Second, the composition of Russian training corpora may have shifted materially since 2024, smoothing the asymmetries Khandelwal et al. and Zhong et al. recorded. Third, the registers represented in Russian-language corpora may differ systematically from English ones: if Russian data are weighted toward literary, scientific, and publicistic texts, with less of the informal hostile discourse abundant in English web corpora, lower rates of bad-form and normalization harms would follow independently of any alignment-transfer mechanism. The present design cannot separate these accounts; all three, however, locate the effect in the production pipeline rather than in any property of Russian-speaking users.

Seven limitations qualify these results. First, the intra-class correlation of vignette is 0.13 on both composites: roughly 13% of variance is attributable to scenario formulation, which the random intercepts absorb only in part. Second, expert–expert reliability for empathy ($\alpha = 0.549$) and de-escalation ($\alpha = 0.586$) is lower than for the principal procedural criteria, and conclusions on these two scales are indicative. Third, Gemini served as both a tested therapist model and the production judge; LLMs tend to score outputs structurally similar to their own more favorably, likely inflating Gemini's therapist scores. Notably, any such inflation did not alter the reported hierarchy—Gemini nonetheless placed last on procedural quality—so within-family bias can affect magnitudes but not the ranking. A related caveat is that the judge was validated on dialogues drawn from the same reserved subsample used to develop the guide, which may render the reported judge–expert agreement somewhat optimistic. Fourth, an LLM patient is not equivalent to a person in distress: it holds to its vignette and does not resist the system prompt or exit the conversation for reasons unspecified in its instructions. The source material adds a further layer—the vignettes derive from public forum posts addressed to other people, not from users' actual crisis conversations with LLMs, so the approximation to real help-seeking lies in the lay language, not in the interactional context. Fifth, the 30-round cap, while sufficient for the crisis-handling question (Section 2.3), excludes long-horizon support scenarios in which model behavior may change as context accumulates and a quasi-alliance forms; the relationship between dialogue length and the quality and harm criteria was not examined here. Sixth, each therapist bot received a minimal counselor prompt without explicit safety instructions, matching how users spontaneously consult general-purpose systems; these results therefore characterize off-the-shelf model behavior, not specialized therapeutic chatbots, whose safety filters and interaction architecture are purpose-built, and comparing the two requires a separate design. Seventh, explicit demographic labeling is a coarse manipulation: the null race and gender findings do not rule out bias under implicit sociolinguistic cues, which remains a separate test.

5 Conclusion

This study contributes to a safety-evaluation pipeline for LLMs in psychological support that combines Reddit-derived user-language vignettes with full-factorial bot-to-bot simulation across five general-purpose models, four immediate-risk categories, eight demographic profiles, and two languages. Across 5,753 analyzed dialogues, none of the models demonstrated consistently adequate handling of crisis situations. Response quality was highest for suicidal ideation and lowest for aggression toward others; Claude Opus 4.5 led both composite scales, with the model effect on safety roughly an order of magnitude larger than the largest demographic effect; sociodemographic predictors explained negligible variance, with the exception of a profession effect on harm; and Russian-language dialogues outperformed English on both composites.

Two practical implications follow. Regulatory effort at the model level—mandatory safety thresholds for general-purpose models deployed in psychological-support contexts—will outperform effort at the interaction level, since model choice dominates demographics, language, and crisis category as a determinant of safety. And safety evaluation should use stimuli grounded in user language alongside clinical criteria: response quality diverges sharply by category on user-language vignettes, a pattern that, given prior vignette-based results (Moore et al., 2025), top-down diagnostic stimuli are unlikely to reveal. The profession finding carries a further implication for the professional community itself: as psychologists increasingly use general-purpose models for reflection and informal supervision, markers of their own clinical fluency may weaken precisely the protective responses these systems extend to lay users.

Four directions remain open: multi-model judge panels with aggregated scoring, to remove the within-family judge bias noted in the limitations; validation of the simulation findings against anonymized records of real user–LLM crisis conversations from deployed systems, where such data can be ethically obtained; expanded language coverage beyond the Russian–English pair, to test whether the safety-alignment-transfer interpretation extends to typologically distinct languages; and analysis of dialogue length—both anchoring the cap to deployed-system usage data and testing whether longer dialogues are associated with anchoring and anthropomorphization rather than with better outcomes.

Acknowledgments. The research was supported by the Russian Science Foundation (RSF) (project No. 25-28-0317).

Disclosure of Interests. The authors have no competing interests to declare that are relevant to the content of this article.

References

1. Bourdieu, P. (1991). *Language and symbolic power* (G. Raymond & M. Adamson, Trans.; J. B. Thompson, Ed.). Harvard University Press.

2. Couldry, N., & Mejias, U. A. (2019). *The costs of connection: How data is colonizing human life and appropriating it for capitalism*. Stanford University Press.
3. Feinstein, A. R., & Cicchetti, D. V. (1990). High agreement but low kappa: I. The problems of two paradoxes. *Journal of Clinical Epidemiology*, 43(6), 543–549. [https://doi.org/10.1016/0895-4356\(90\)90158-L](https://doi.org/10.1016/0895-4356(90)90158-L)
4. Fiske, A., Henningsen, P., & Buyx, A. (2019). Your robot therapist will see you now: Ethical implications of embodied artificial intelligence in psychiatry, psychology, and psychotherapy. *Journal of Medical Internet Research*, 21(5), Article e13216. <https://doi.org/10.2196/13216>
5. Gabriel, S., Puri, I., Xu, X., Malgaroli, M., & Ghassemi, M. (2024). Can AI relate: Testing large language model response for mental health support. In Y. Al-Onaizan, M. Bansal, & Y.-N. Chen (Eds.), *Findings of the Association for Computational Linguistics: EMNLP 2024* (pp. 2643–2660). Association for Computational Linguistics. <https://doi.org/10.48550/arXiv.2405.12021>
6. Gwet, K. L. (2008). Computing inter-rater reliability and its variance in the presence of high agreement. *British Journal of Mathematical and Statistical Psychology*, 61(1), 29–48. <https://doi.org/10.1348/000711006X126600>
7. Khandelwal, A., Agarwal, U., Tanmay, K., & Choudhury, M. (2024). Do moral judgment and reasoning capability of LLMs change with language? A study using the multilingual Defining Issues Test. In Y. Graham & M. Purver (Eds.), *Proceedings of the 18th Conference of the European Chapter of the Association for Computational Linguistics (Vol. 1: Long papers)* (pp. 2882–2894). Association for Computational Linguistics. <https://aclanthology.org/2024.eacl-long.176/>
8. Kleinman, A. (1988). *The illness narratives: Suffering, healing, and the human condition*. Basic Books.
9. Landis, J. R., & Koch, G. G. (1977). The measurement of observer agreement for categorical data. *Biometrics*, 33(1), 159–174. <https://doi.org/10.2307/2529310>
10. Low, D. M., Rumker, L., Talkar, T., Torous, J., Cecchi, G., & Ghosh, S. S. (2020). Natural language processing reveals vulnerable mental health support groups and heightened health anxiety on Reddit during COVID-19: Observational study. *Journal of Medical Internet Research*, 22(10), Article e22635. <https://doi.org/10.2196/22635>
11. Moore, J., Grabb, D., Agnew, W., Klyman, K., Chancellor, S., Ong, D. C., et al. (2025). Expressing stigma and inappropriate responses prevents LLMs from safely replacing mental health providers. In Proceedings of the 2025 ACM Conference on Fairness, Accountability, and Transparency (pp. 599–627). Association for Computing Machinery. <https://doi.org/10.1145/3715275.3732039>
12. Pendse, S. R., Jain, M., Kumar, N., & De Choudhury, M. (2025). Implicit gender, racial, and ethnic biases in large language models: An audit study of automated psychiatric diagnoses. SSRN. <https://doi.org/10.2139/ssrn.5317730>
13. Poulain, R., Adiba, F. I., Fayyaz, H., & Beheshti, R. (2026). Bias Patterns in the Application of LLMs for Clinical Decision Support: A Comprehensive Study. *Delaware journal of public health*, 12(1), 54–67. <https://doi.org/10.32481/djph.2026.03.10>
14. Rousmaniere, T., Zhang, Y., Li, X., & Shah, S. (2025). Large language models as mental health resources: Patterns of use in the United States. *Practice Innovations*, 10(3), 175–186. <https://doi.org/10.1037/pri0000292>
15. Song, I., Pendse, S.R., Kumar, N., & de Choudhury, M. (2025). The Typing Cure: Experiences with Large Language Model Chatbots for Mental Health Support. *Proceedings of the ACM on Human-Computer Interaction*, 9, 1 - 29. <https://doi.org/10.1145/3757430>

16. Steenstra, I., & Bickmore, T. (2025). A risk ontology for evaluating AI-powered psychotherapy virtual agents. In *Proceedings of the 25th ACM International Conference on Intelligent Virtual Agents* (Article 12, pp. 1–10). Association for Computing Machinery. <https://doi.org/10.1145/3717511.3749286>
17. Tamoyan, H., Schuff, H., & Gurevych, I. (2025). LLM roleplay: Simulating human-chatbot interaction. In *Proceedings of the Third Workshop on Social Influence in Conversations (SICoN 2025)* (pp. 1–26). Association for Computational Linguistics. <https://aclanthology.org/2025.sicon-1.1/>
18. Yeo, Y. H., Peng, Y., Mehra, M., Samaan, J., Hakimian, J., Clark, A., et al. (2025). Evaluating for evidence of sociodemographic bias in conversational AI for mental health support. *Cyberpsychology, Behavior, and Social Networking*, 28(1), 44–51. <https://doi.org/10.1089/cyber.2024.0199>
19. Zhong, Q., Yun, Y., & Sun, A. (2024). *Cultural value differences of LLMs: Prompt, language, and model size*(arXiv:2407.16891). arXiv. <https://doi.org/10.48550/arXiv.2407.16891>